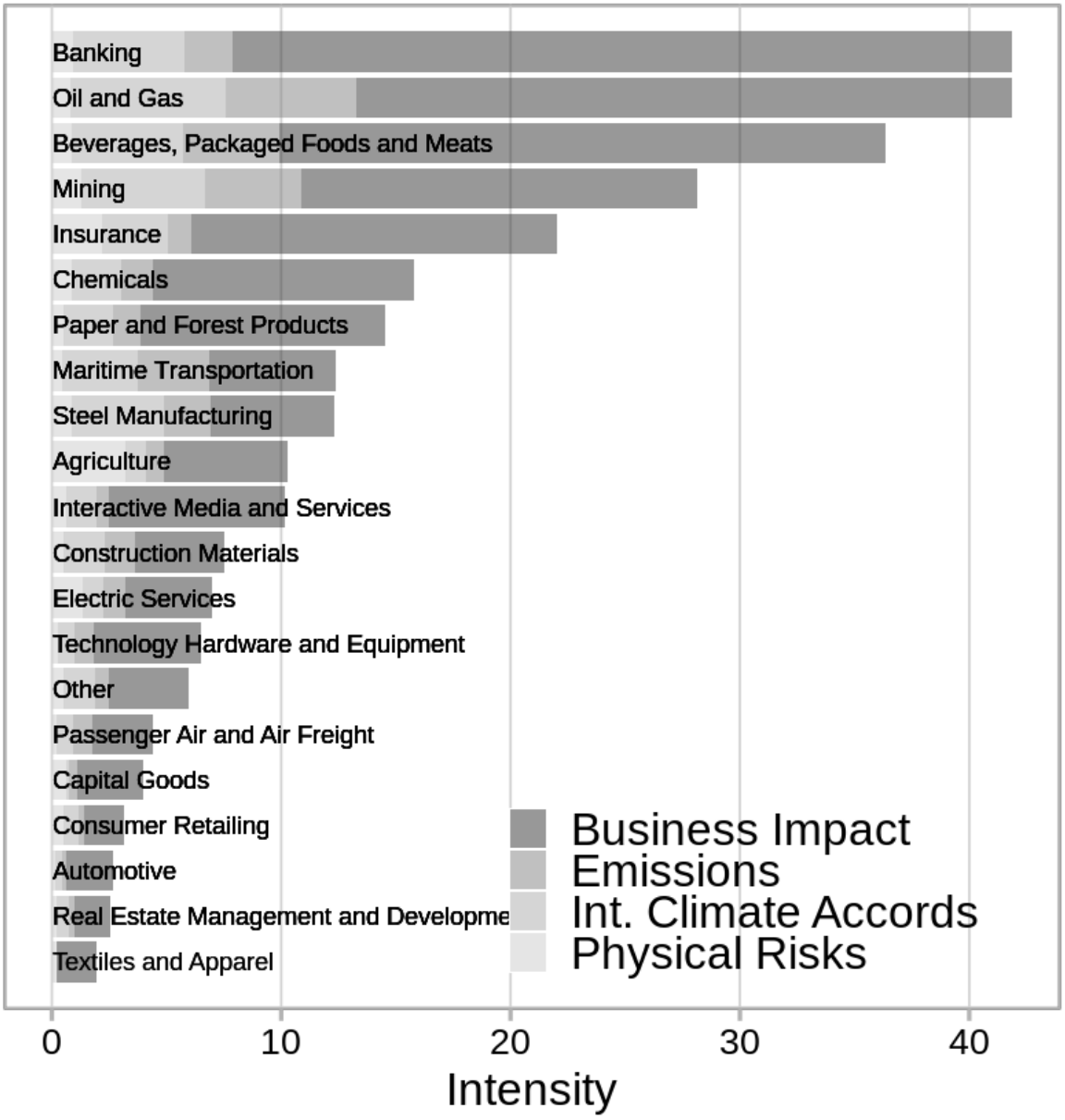
Figure 4. Clustered Intensity by Industry for Foreign Filings (Form 20-F)



This figure presents the analysis of Form 20-F annual reports submitted to the Commission between Jan. 1, 2021, and Dec. 31, 2022. We exclude any Form 20-Fs that were not annual reports. NLP was used to analyze sentences contained within the annual filings and classify them into four broad topics (i.e., clusters): business impact, emissions, international climate accords, and physical risks. Intensity refers to the average number of relevant sentences per registrant, which is calculated by taking the aggregate number of sentences in an industry and dividing it by the total number of registrants within the industry.

The Commission staff's findings are consistent with one academic study that looked at the extent of climate-related disclosures by Commission registrants.[2642] In this study, a review of Form 10-K filings from Russell 3000 companies over the last 12 years found that the majority of climate-related disclosure is focused on transition risks,[2643] consistent with the above Commission staff analysis that finds that annual filings contain more discussion on emissions and international climate accords relative to physical risks. This study further found that while 35 percent of Russell 3000 Index companies provided climate-related information in 2009, this figure grew to 60 percent in 2020.[2644] The study also found that the extent of disclosure for a given report has increased.[2645] In 2009, companies mentioned climate risks 8.4 times on average in their Form 10-K.[2646] This figure grew to 19.1 times in 2020.[2647]

The Proposing Release included a similar analysis of climate-related disclosures in Commission filings using data from earlier years.[2648] That analysis also found that filings by registrants in the electric services and oil and gas industries have the most robust climate-related discussions.[2649] In response to this finding, one commenter suggested that the current "principles-based approach is working successfully, as these are industries where climate-related

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2642 *See* Parker Bolstad, Sadie Frank, Erick Gesick & David Victor, *Flying Blind: What Do Investors Really Know About Climate Change Risks in the U.S. Equity and Municipal Debt Markets* (Hutchins Center Working Paper 67, 2020) ("Hutchins Center Working Paper").

2643 *See id.*

2644 *See id.* The methodology uses a series of keywords to determine whether a company provides climate-related disclosures. Some keywords may occur in non-climate contexts, which the authors note may introduce some bias into the statistics.

2645 *See id.*

2646 *See id.*

2647 *See id.*

2648 *See* Proposing Release, section IV.A.5.a.

2649 *See id*.

factors are more likely to have a material impact on the present value of future cash flows."[2650] We disagree. The Commission staff's analysis focuses on the incidence of climate-related discussion in annual reports (Forms 10-K and 20-F).[2651] The fact that the incidence of disclosures may be correlated with the likelihood that climate-related risks are material to a particular company does not demonstrate that registrants are fully disclosing their material climate-related risks to investors. For instance, registrants may strategically omit information that could be perceived as negative or adverse,[2652] and some studies point to the potential for substantial underreporting of material climate-related information within the current principles-based reporting regime.[2653]

In addition, one commenter suggested the Commission examine analyst reports and interactions involving analysts to assess "the significance of ESG factors relative to other factors for determining the value of securities."[2654] There is academic research that considers analyst reports; this literature has found that, while very few analyst reports traditionally discuss topics related to climate, climate-related disclosures can offer useful predictive signals about future financial performance for firms whose industries are most exposed to climate-related risk and

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2650 *See* Overdahl exhibit to letter from Chamber.

2651 *See also* section IV.A.5 for an update of the analysis in the Proposing Release.

2652 A recent analysis, for example, showed that absent mandatory requirements from regulators, voluntary disclosures following third-party frameworks were generally of poor quality and that companies making these disclosures cherry-picked to report primarily non-material climate risk information. *See* Julia Bingler, Mathias Kraus, Markus Leippold & Nicolas Webersinke, *Cheap Talk and Cherry-Picking: What ClimateBert Has to Say on Corporate Climate Risk Disclosures,* 47 Fin. Rsch. Letters, Article 102776 (June 2022) ("Bingler *et al.*") (reviewing annual reports for fiscal years 2014-2019 – i.e., before and after the introduction of TCFD recommendations – for a sample of 818 TCFD-supporting firms).

2653 Lee Reiners & Charlie Wowk, *Climate Risk Disclosures & Practices* (2021), available at https://econ.duke.edu/sites/econ.duke.edu/files/documents/Climate-Risk-Disclosures-and-Practices.pdf; Bingler *et al.*; Morningstar, *Corporate Sustainability Disclosures* (2021), available at https://www.morningstar.com/en-uk/lp/corporate-sustainability-disclosures ("Companies will disclose the good and hide the bad while disclosure remains voluntary.").

2654 *See* Overdahl exhibit to letter from Chamber.

can influence analysts to revise their target prices for these firms.[2655] Other research has found that Form 10-K disclosures on material climate risks are associated with increased precision and lower dispersion in analysts' earnings forecasts.[2656] Similarly, in the context of earnings conference calls involving analysts, discussions concerning exposure to climate-related risks have been shown to contain important information that is priced in stocks and options.[2657] Relatedly, the same commenter suggested that the Commission conduct an event study to study price or volume responses to climate-related disclosures.[2658] We decline to follow the suggestion in light of the support in peer reviewed literature for the importance of climate-related disclosures to investors.[2659] Existing research finds an increase in stock price volatility around the day when GHG or carbon emissions are disclosed in a Form 8-K filing.[2660] This suggests that investors find such disclosures to be informative.

2655 *See* Jesse Yuen-Fu Chan, *Climate Change Information and Analyst Expectations* (July 29, 2022) (Ph.D. dissertation, University of Texas, Austin), available at https://repositories.lib.utexas.edu/items/092f6e82-c4b1-4d61-a83b-207643cbb62d.

2656 *See* Walid Ben-Amar *et al.*, *Do Climate Risk Disclosures Matter to Financial Analysts?*, J. of Bus. Fin. & Acct. (2023), available at https://onlinelibrary.wiley.com/doi/10.1111/jbfa.12778 (using the Materiality Map provided by the Sustainability Accounting Standards Board (SASB) to show that the association between improvements to forecast precision and climate risk disclosure is present only when climate risk is deemed financially material at the industry level according to SASB).

2657 *See* Zacharias Sautner, *et al.*, *Firm-Level Climate Change Exposure*, 78 J. of Fin. 1449 (Feb. 2023) ("Sautner, *et al.* (2023)"); Qing Li, Hongyu Shan, Yuehua Tang & Vincent Yao, *Corporate Climate Risk: Measurement and Responses* (forthcoming Rev. Fin. Stud., 2020), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3508497 (retrieved from SSRN Elsevier database); additionally, researchers have noted that, although the frequency of such climate-related discussions have historically been low, there has been an increase in recent years. *See* Michał Dzieliński, Florian Eugster, Emma Sjöström & Alexander F. Wagner, *Climate Talk in Corporate Earnings Calls* (Swiss Fin. Inst. Rsch. Paper Series 22-14, 2022), available at https://ideas.repec.org/p/chf/rpseri/rp2214.html.

2658 *See* Overdahl exhibit to letter from Chamber.

2659 *See also* discussions in sections IV.B.1 and IV.C.1.a.

2660 *See* Paul A. Griffin, David H. Lont & Estelle Y. Sun, *The Relevance to Investors of Greenhouse Gas Emission Disclosures*, 34 Contemp. Acct. Rsch. 1265 (2017).

b. Additional Trends in Climate-Related Disclosures

As discussed below, a number of industry and advocacy groups have examined the scope of voluntary climate-related disclosures, and their findings are relevant to assess the economic impact of the final rules.

i. Prevalence and Scope of Climate-Related Disclosures

As discussed in the Proposing Release,[2661] one organization, in collaboration with several other organizations, conducted a survey of a sample of 436 U.S. public companies across 17 industries that range from small to large in terms of market capitalization.[2662] According to the survey, over half of the companies (52 percent) published a CSR, sustainability, or a similar report, the contents of which commonly include information regarding climate-related risks. The most frequently discussed topics in such reports were energy (74 percent), emissions (70 percent), environmental policy (69 percent), water (59 percent), climate mitigation strategy (57 percent), and supplier environmental policies (35 percent). Among the registrants that reported climate-related information to the public, the majority disclosed such information via external reports or company websites rather than through regulatory filings. Similar to the Commission staff's review, the survey found that about a third (34 percent) of the respondents disclosed information regarding "risks related to climate change, greenhouse gas emissions, or energy sourcing" in their Commission filings.[2663] Among these companies, 82 percent disclosed such

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2661 *See* Proposing Release, at section IV.A.5.b.

2662 *See* Center for Capital Markets, *2021 Survey Report: Climate Change & ESG Reporting from the Public Company Perspective*, available at https://www.centerforcapitalmarkets.com/wp-content/uploads/2021/08/CCMC_ESG_Report_v4.pdf. Sixty-seven percent of survey respondents have market capitalization below $5 billion, while 32% are below $700 million.

2663 *See id.*

information in Risk Factors, 26 percent in MD&A, 19 percent in the Description of Business, and 4 percent in Legal Proceedings.[2664]

One institute issues annual analyses of sustainability reports by the companies belonging to the Russell 1000 Index.[2665] The institute found that in calendar year 2022, a record high of 90 percent of these companies published sustainability reports, which commonly include climate-related information—up from 60 percent in 2018.[2666] In particular, sustainability reporting reached an all-time high of 98 percent for companies in the top half of the Russell 1000 Index (which roughly comprises the S&P 500 Index). However, the most significant change was among companies in the bottom half of the Russell 1000 Index, where sustainability reporting percentage increased to 82 percent, up from 34 percent in 2018. The percentage of companies from each Global Industry Classification Standard ("GICS") sector[2667] that published a sustainability report in 2021 were: Communications (56 percent), Consumer Discretionary (81 percent), Consumer Staples (91 percent), Energy (94 percent), Financials (85 percent), Health Care (69 percent), Industrials (89 percent), Information Technology (71 percent), Materials (95 percent), Real Estate (90 percent), and Utilities (100 percent).

2664 *See id.*

2665 *See* G&A, *Sustainability Reporting Trends*, available at https://www.ga-institute.com/research/ga-research-directory/sustainability-reporting-trends; *see also* Proposing Release, section IV.A.5.b.

2666 *See* G&A, *2023 Sustainability Reporting in Focus*, available at https://www.ga-institute.com/research/ga-research-directory/sustainability-reporting-trends/2023-sustainability-reporting-in-focus.html; *see also* past reports, available at https://www.ga-institute.com/research/ga-research-directory/sustainability-reporting-trends.html.

2667 For more information on GICS sector categories, *see* MSCI, The Global Industry Classification Standard (GICS), available at https://www.msci.com/our-solutions/indexes/gics (last visited Feb. 28, 2024).

Notwithstanding these investor-led initiatives, disclosures currently vary considerably in terms of coverage, location, and presentation across companies,[2668] making it difficult for investors to navigate through different information sources and filings to identify, compare, and analyze climate-related information.[2669] For example, one commenter submitted a survey reporting that institutional investors spend an average of $257,000 and $357,000 on "collecting climate data related to assets" and "internal climate-related investment analysis," respectively.[2670] An academic study similarly finds that "there exists considerable heterogeneity in what and how firms report about their CSR activities…The heterogeneity in reported CSR topics makes it difficult for users to compare disclosures and to benchmark firms' underlying CSR performance."[2671] Some studies and commenters have asserted that current disclosures are

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2668 *See*, *e.g.,* TCFD Report, *supra* note 46, at 16; *see also* IOSCO Report, *supra* note 1089; GAO, *Climate-Related Risks* (2018), available at https://www.gao.gov/assets/gao-18-188.pdf (reporting that "investors may find it difficult to navigate through the filings to identify, compare, and analyze the climate-related disclosures across filings"); letter from Bloomberg.

2669 *See* letters from Calvert ("Calvert purchases third party vendor data to support our ability to assess companies on their ESG factors and that provide specific data related to climate change, where available. Often vendor information is estimated when a company has not disclosed information on its climate-related risks. Sometimes the estimates are made across industries, based on what other more proactive peers have disclosed. We are concerned about the lack of accuracy fostered by estimation methodologies, and also the trend for these methodologies to under-estimate actual emissions."); Boston Trust Walden ("our analysts examine quantitative and qualitative climate-related corporate disclosure to enhance our understanding of the existing and potential financial outcomes associated, ranging from risks (e.g., losing the license to operate) to opportunities (e.g., generating new sources of revenue). In the absence of mandated disclosure requirements, we rely on the data of third-party research providers, which includes a mix of issuer provided data and estimates. Our analysts then seek to fill data gaps through additional research and analysis, outreach via written requests, meetings, and shareholder resolutions seeking the expanded disclosure we require. These processes for gathering necessary climate-related disclosures are inefficient and resource intensive."); NY Office of the State Comptroller; and State of Vermont Pension Investment Commission.

2670 *See* ERM survey attached to letter from ERM (June 16, 2022) ("ERM survey").

2671 *See* Hans B. Christensen, Luzi Hail & Christian Leuz, *Mandatory CSR and Sustainability Reporting: Economic Analysis and Literature Review,* 26 Rev. of Acct. Stud. 1176 (2021) ("Christensen *et al.* (2021)") at 1194.

often vague and boilerplate, creating challenges for investors.[2672] Industry observers and some commenters also report that many registrants that currently provide voluntary climate-related disclosures through sustainability reports often take longer than 12 months after their fiscal year end to disclose decision-relevant data, raising concerns about the timeliness of these reports for investors.[2673] As noted in section II.A.2, many commenters stated that the Commission's current reporting requirements do not yield adequate or sufficient information regarding climate-related risks.[2674]

ii. GHG Emissions Reporting

Commission staff also analyzed the number of registrants that recently reported Scope 1 and 2 emissions data. In this analysis, Commission staff utilized a database that compiles emissions data (among other ESG-related information) from companies' annual filings, sustainability reports, or other public disclosures.[2675] The number of registrants that are covered in

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[2672] *See* SASB, The State of Disclosure: An Analysis of the Effectiveness of Sustainability Disclosure in SEC Filing*s* (2017), available at https://www.sasb.org/wp-content/uploads/2019/08/StateofDisclosure-Report-web112717.pdf (reporting that about 50% of Commission registrants provide generic or boilerplate sustainability information in their regulatory filings); *see also* letter from The Institute for Policy Integrity (Policy Integrity) at New York University School of Law, 1 Environmental Defense Fund (EDF), and Professor Madison Condon https://www.sec.gov/comments/s7-10-22/s71022-20131770-302204.pdf ("Existing disclosure regulations and guidance have proved insufficient to address this asymmetry. In a 2020 study of climate risk disclosures in 10-K filings, the Brookings Institution concluded that though '[d]isclosure has risen sharply,' '[m]ore firms are disclosing more general information that is essentially of no utility to the marketplace.'").

[2673] *See supra* note 2570; *see also* letter from Calvert ("Last year, when evaluating disclosure rates of companies in our equities portfolios, we found 57% of 2,207 companies disclosed their Scope 1 and 2 emissions with a one to two year delay . . . [B]y the time this data is gathered, there may be a long lag time to the point of disclosure—it is not uncommon that GHG emissions disclosure is already 12-18 months out of date once it is actually published.").

[2674] *See supra* section II.A.2.

[2675] Commission staff used the Refinitiv ESG database, which covers over 88% of global market capitalization, across more than 700 different ESG metrics. The U.S. coverage broadly includes listed companies belonging to the Russell 3000 Index. The emissions data used in this analysis was extracted from Refinitiv on Feb 11, 2024. *See* Refinitiv, *Environmental, Social And Governance Scores From Refinitiv* (May 2022), available at https://www.lseg.com/content/dam/marketing/en_us/documents/methodology/refinitiv-esg-scores-methodology.pdf

this database is 5,535, which comprises the matched sample. From this matched sample, about 20 percent of registrants (1,125 out of 5,535) reported their Scope 1 and Scope 2 emissions in fiscal year 2021, with the highest disclosure rate found among LAFs (50 percent).[2676] In fiscal year 2022, about 18% of registrants (870 out of 5,535) reported their Scope 1 and 2 emissions, with the disclosure rate among LAFs at 42 percent. These and other statistics are presented in Table 5.

Table 5. Number of registrants that disclose Scope 1 and 2 emissions using third-party data[1]

				Scope 1 and 2 emissions disclosures			
				FY 2021		FY 2022[6]	
Filer status (1)	SEC registrants[2] (2)	Registrants covered in third-party database[3] (3)	Coverage rate[4] (4)	Disclosed (5)	Disclosure rate[5] (6)	Disclosed (7)	Disclosure rate[7] (8)
LAF	2,528	2,059	81%	1,026	50%	870	42%
AF	444	334	75%	57	17%	50	15%
NAF	507	154	30%	8	5%	15	10%
SRC/EGC	4,265	2,988	70%	34	1%	50	2%
Total	7,744	5,535	71%	1,125	20%	985	18%

1. Commission staff used the Refinitiv ESG database. *See supra* note 2675.
2. These statistics are based on SEC registrants filing annual reports in calendar year 2022. *See supra* note 2578. For LAF, AF, and NAF registrant counts, only those that are not SRCs or EGCs are included. We note that several non-SRC/EGC registrants did not disclose their filer status, thus the total registrant count in Table 5 is not the same as what is indicated in section IV.A.1.
3. The matched sample consists of the number of registrants that are covered in the Refinitiv ESG database.
4. Column (4) = (Column (3))/(Column (2)).
5. Column (6) = (Column (5))/(Column (3)).
6. Data collection of GHG emissions disclosure can lag by 18 months or longer. As a result, the number of disclosers for FY 2022 may not be complete and thus understated.
7. Column (8) = (Column (7))/(Column (3)).

[2676] These percentages may be understated to the extent that Refinitiv may not be able to fully track all emissions disclosures made by Commission registrants. Conversely, compared to the full sample of Commission registrants, these figures may be overstated given that smaller firms outside of Refinitiv's coverage universe (i.e., those outside of the Russell 3000) are less likely to report emissions.

We note that the number of registrants providing disclosure in fiscal year 2022 may be understated given that data collection of GHG emissions can lag by up to 18 months.[2677] To estimate how this number could potentially increase upon the completion of data collection, we consider the following assumption: for those registrants that disclosed only in fiscal year 2021 (but not in fiscal year 2022), we assume that their fiscal year 2022 disclosures are forthcoming.[2678] Within the matched sample, there are 263 LAFs that disclosed in fiscal year 2021 but not in fiscal year 2022. The corresponding number for AFs is 17. If we assume that these registrants subsequently provide their fiscal year 2022 disclosures, the fiscal year 2022 disclosure rate for LAFs would increase from 42% to 55%[2679] and that of AFs would increase from 15% to 20%.[2680] We recognize, however, that the above assumption may not hold true for all of these registrants.

Commission staff also analyzed U.S. companies that voluntarily responded to CDP's questionnaire and publicly disclosed their responses.[2681] In 2022, 1,311 domestic companies

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2677 The Commission understands that data collection of GHG emissions for FY 2022 is ongoing. In addition, some industry observers have noted that "many companies still take more than 12 months after their fiscal year to disclose their sustainability data," *see, e.g.*, Corporate Knights, *supra* note 2570; letter from Morningstar ("Currently, a lack of clear disclosure standards for the timing of 'sustainability reports,' which is the primary source for emissions data, greatly hinders investor knowledge. For example, some registrants released 2021 reports—detailing 2020 data—as late as November 2021.")

2678 These registrants have demonstrated that they have Scope 1 and 2 emissions measurement and disclosure processes in place. It is therefore plausible that they have forthcoming disclosures for FY 2022 that is not yet in the dataset.

2679 (870 + 263)/2059 = 55%.

2680 (50+17)/334 = 20%.

2681 This analysis is based on data provided to the Commission from CDP, available at https://www.sec.gov/comments/s7-10-22/s71022-206599-416182.xlsx. CDP operates a global disclosure system that enables companies, cities, states and regions to measure and manage their environmental risks, opportunities and impacts. Despite not being a framework like GRI, SASB and TCFD, CDP's questionnaires gather both qualitative and quantitative information from across governance, strategy, risk,